ONLINE QUIZ SYSTEM

A PROJECT REPORT

Submitted by

JEYASRI G (8115U25AD040)

in partial fulfillment of requirements for the award of the course

CGB1122 – DATA STRUCTURES

in

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Under the guidance of

ASHA G

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY 621112

ANNA UNIVERSITY

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SAMAYAPURAM – 621 112

BONAFIDE CERTIFICATE

Certified that this project report titled **“ONLINE QUIZ SYSTEM”** is the bonafide work of **JEYASRI G(8115U25AD040)** who carried out the project work under my supervision.

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DECLARATION BY THE CANDIDATE

I jointly declare that the project report on **"INTEGRATED ONLINE MEDICINE ODERING AND HEALTHCARE HUB BY USING PHP (MEDILINK)"** is the result of original work done by us and best of our knowledge. This project report is submitted on the partial fulfillment of the requirement of the award of the course **CGB1121- DATA STRUCTURES.**

Signature

JEYASRI G

Place: Samayapuram

Date:

ACKNOWLEDGEMENT

I thank the almighty GOD, without whom it would not have been possible for us to complete our project. I wish to address our profound gratitude to **Dr.K.RAMAKRISHNAN**, Chairman, K. Ramakrishnan College of Engineering (Autonomous), who encouraged and gave us all help throughout the course.

I extend my hearty gratitude and thanks to our honorable and grateful Executive Director **Dr.S.KUPPUSAMY, B.Sc., MBA., Ph.D.,** K. Ramakrishnan College of Engineering (Autonomous).

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I render our sincere thanks to the Course Coordinator and other staff members for providing valuable information during the course.

INSTITUTE VISION AND MISSION

VISION OF THE INSTITUTE:

To achieve a prominent position among the top technical institutions.

MISSION OF THE INSTIITUTE:

M1: To bestow standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.

M2: To nurture research and entrepreneurial skills among students in cutting edge technologies.

M3: To provide education for developing high-quality professionals to transform the society.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations

4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations

6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice

7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The Online Quiz System is a computer-based application developed to conduct quizzes in an efficient and automated manner. It helps users answer multiple-choice questions and receive scores instantly. The system reduces manual checking, saves time, and improves accuracy in result calculation. It provides an easy and interactive platform for students and teachers to participate in quiz activities.This project helps save time, improves accuracy, and minimizes human errors during evaluation. It offers a user-friendly interface so that users can easily read questions, select answers, and submit the quiz without difficulty. The system can be used by students for self-assessment and by teachers for conducting tests in a convenient way.

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LIST OF ABBREVIATIONS

ABBREVIATIONS

DLL	-	Doubly Linked List
UI	-	User Interface
CLI	-	Command Line Interface
GUI	-	Graphical User Interface
NLP	-	Natural Language Processing
APIs	-	Application Programming Interface

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In the modern education system, online learning tools play an important role. Traditional quiz methods require paper, manual evaluation, and more time. The Online Quiz System is designed to overcome these issues by providing a digital platform where quizzes can be conducted easily. It allows users to answer questions, submit responses, and get results immediately.

1.2 PURPOSE AND IMPORTANCE

The main purpose of the Online Quiz System is to provide a simple and efficient way of conducting quizzes through a computer-based platform. It replaces the traditional paper-based quiz method and allows users to answer questions digitally. This makes the quiz process faster, easier, and more convenient for both students and teachers.

Another important purpose of this system is to reduce manual effort in checking answers and calculating marks. In the traditional method, teachers need more time to evaluate answer sheets. With the Online Quiz System, scores are calculated automatically, which saves time and provides accurate results instantly.

The importance of this project also lies in promoting digital learning and smart education methods. It helps students practice tests anytime and improve their knowledge through regular assessments. Instant feedback allows users to identify mistakes and learn from them quickly.

This system is also important because it is cost-effective and eco-friendly. It reduces the use of paper, printing expenses, and physical storage of records. The Online Quiz System is a modern solution that improves efficiency, accuracy, and user experience in the examination process.

1.3 OBJECTIVES

1.To conduct quizzes digitally.

2.To calculate scores automatically.

3.To reduce manual effort.

4.To save time and improve accuracy.

5.To provide instant results.

1.4 PROJECT SUMMARIZATION

The Online Quiz System is a simple and effective application developed to conduct quizzes in a digital format. It allows users to answer multiple-choice questions and receive instant scores after completion. The project helps reduce manual work involved in checking answers and calculating marks. It improves accuracy, saves time, and provides a user-friendly experience. This system is useful for schools, colleges, and training centers, and it supports modern learning methods through quick and organized assessment processes.

Overview:

The Online Quiz System is a computer-based application designed to conduct quizzes in a simple, fast, and organized manner. It provides an easy platform for users to attend quizzes digitally without using paper-based methods.

Main Function:

The system displays multiple-choice questions to users, accepts their answers, and evaluates the responses automatically.

Educational Use:

Students can test their knowledge regularly and improve their performance through practice.

Future Scope:

The project can be enhanced by adding features such as timer-based quizzes, random questions, user login, database storage, and leaderboard systems.

CHAPTER 2

PROJECT METHODOLOGY

2.1 INTRODUCTION TO SYSTEM ARCHITECTURE

The system architecture of the Online Quiz System represents the overall structure and working flow of the application. It shows how different modules are connected to perform the quiz process smoothly and efficiently. The architecture helps in understanding how user input is processed and how results are generated.

2.1.1 High-Level System Architecture

The High Level System Architecture of the Online Quiz System describes the major components of the system and the interaction between them. It provides a clear overview of how the application functions from user login to result generation. The architecture is designed to ensure smooth data flow, easy management, and efficient quiz processing.

2.1.2 Components of the System Architecture

The Online Quiz System consists of several important components that work together to perform the quiz process effectively. Each component has a specific role in handling user interaction, quiz management, evaluation, and result generation.

1. User Login Module

This component allows students or users to enter the system using their login credentials. It provides secure access to the quiz application.

2. Quiz Module

The Quiz Module is responsible for displaying questions and options to the user.

It manages the flow of the quiz and allows users to submit their answers.

3. Question Bank

This component stores all quiz questions along with answer choices. Questions are retrieved from the database or stored records when the quiz starts.

4. Evaluation Module

The Evaluation Module checks the answers submitted by the user and compares them with the correct answers. It ensures accurate assessment of performance.

5. Score Calculation Module

This component calculates the total marks based on correct answers. It automatically generates the final score without manual effort.

6. Result Storage Module

The Result Storage Module saves user scores and quiz details for future reference. It helps maintain records of student performance.

7. Display Result Module

This component shows the final result, score, and performance summary to the user after quiz completion.

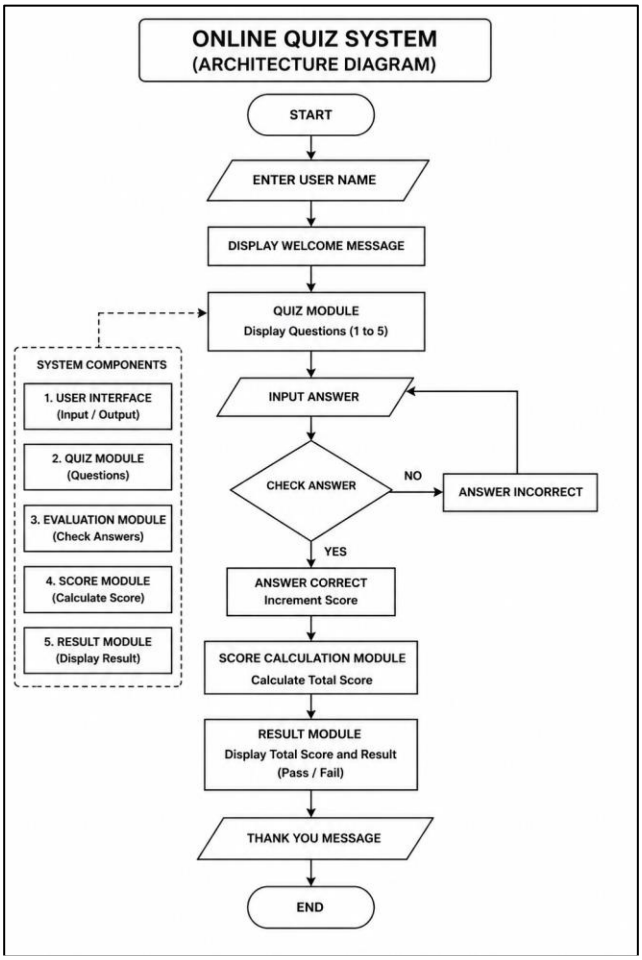
8. End Process

After displaying the result, the quiz session ends successfully.

2.2 DETAILED SYSTEM ARCHITECTURE DIAGRAM

The system architecture of the Online Quiz System represents the overall structure and working flow of the application. It shows how different modules are connected to perform the quiz process smoothly and efficiently. The architecture helps in understanding how user input is processed and how results are generated.

Figure 2.1 : Architecture Diagram (Sample)



CHAPTER 3

DATA STRUCTURE

PREFERANCE

3.1 JUSTIFICATION FOR SELECTING DATA STRUCTURES

Data structures play an important role in the Online Quiz System because they help organize, store, and process quiz data efficiently. The selection of suitable data structures improves system performance, speed, and accuracy.

1. Array

Arrays are used to store quiz questions, answer options, and correct answers in an ordered manner. They allow easy access to data using index positions.

2. Structure

Structures are preferred to store related data such as question number, question text, options, and correct answer under one unit. This makes data handling simple and organized.

3. String

Strings are used to store question statements, option texts, and user names. They are useful for managing text-based information in the system.

4. Variables

Integer variables are used for score calculation, question count, and option selection. They help in performing arithmetic operations quickly.

5. Why Preferred

These data structures are simple, efficient, and easy to implement in C programming. They provide better memory management and smooth execution for small to medium-sized quiz applications.

3.2 COMPARISON WITH OTHER DATA STRUCTURES

In the Online Quiz System, Arrays, Structures, Strings, and Integer Variables are preferred over other data structures because they are simple, efficient, and easy to implement. Arrays are better than Linked Lists, Stacks, and Queues because they allow direct access to questions and answers using index positions. Structures are more suitable than Unions because they can store all question details separately and access them simultaneously. Strings are more convenient than handling individual characters because they make storing and displaying text easier. Compared to complex data structures, these selected structures require less memory management, reduce coding complexity, and provide faster execution for small to medium-sized quiz applications.

3.3 ADVANTAGES AND DISADVANTAGES

3.3.1. Advantages of Using a Array,structure,string:

The Online Quiz System offers many advantages in conducting quizzes efficiently and accurately. It reduces manual work involved in preparing papers, checking answers, and calculating marks. The system saves time by generating instant results after quiz completion. It improves accuracy by minimizing human errors during evaluation. The user-friendly interface makes it easy for students and teachers to use. It also reduces paper usage, making it cost-effective and eco-friendly. The system helps students practice regularly, improves learning interest, and supports modern digital education methods. Overall, it is a fast, reliable, and convenient solution for quiz management.

3.3.2 Disadvantages of Using a Array,structure,string:

The Online Quiz System also has some disadvantages and limitations. It requires a computer or digital device to access the quiz, which may not be available to all users. Technical issues such as power failure, system crashes, or software errors can interrupt the quiz process. In a basic version, the number of questions and features may be limited. It may also require internet connectivity if developed as an online platform. Some users who are not familiar with computers may face difficulty while using the system. Regular maintenance and updates are needed to keep the system working properly. Overall, while useful, the system depends on technology and proper management.

CHAPTER -4

DATA STRUCTURE METHODOLOGY

4.1 ARRAY DATA STRUCTURE

The Online Quiz System mainly uses the Array Data Structure to store quiz questions, answer options, and correct answers in a sequential manner. Arrays are chosen because they allow quick access to elements using index positions and are easy to manage in quiz applications.

Key features :

- Supports direct access using index numbers.
- Easy to implement and maintain.
- Faster retrieval of stored data.
- Suitable for fixed-size quiz records.

Array operations:

The major operations performed using arrays in this system are initialization, insertion, deletion, searching, and traversal for managing quiz data efficiently.

4.2 STRUCTURE DATA TYPE

Structure is used to group different data items such as question text, options, and correct answer into a single unit. It helps in organizing quiz records clearly and systematically.

Key features:

- Improves data organization.
- Easy handling of question details.
- Reduces code complexity.
- Useful for storing complete quiz records.

Structure operations:

Structures are used for initialization of records, storing question details, updating records, and displaying quiz information when required.

4.3. STRING DATA TYPE

Strings are used to store textual information such as quiz questions, option names, and user details. They are essential for handling character-based data in the system.

Key Features:

- Stores words and sentences.
- Easy display of text data.
- Useful for questions and options.

String operations:

String operations in this system include input, output, comparison, copying, and displaying text-based quiz content

CHAPTER-5

MODULES

5.1 USER MANAGEMENT MODULE

The User Management Module is responsible for handling user access to the Online Quiz System. It allows users to log in with their credentials and enter the application securely. This module ensures that only authorized users can access the quiz and start the examination process. It acts as the entry point of the system .The User Management Module controls the entry and identification process of users in the Online Quiz System. It collects user details such as username and password, verifies them, and grants access to the quiz platform. This module helps maintain security and ensures that each user can use the system properly. It also manages basic user information required for quiz participation.

5.2 QUIZ PROCESSING MODULE

The Quiz Processing Module manages the main quiz activities of the system. It stores quiz questions, displays them one by one to the user, accepts selected answers, and controls the overall flow of the quiz session. This module ensures that the quiz is conducted smoothly and in an organized manner. The Quiz Processing Module is the core part of the Online Quiz System. It is responsible for presenting questions with answer options, recording user responses, and maintaining the sequence of questions throughout the quiz. This module handles all quiz-related operations and ensures that the user can complete the test without interruption. It makes the quiz process simple, smooth, and efficient.

5.3 RESULT MANAGEMENT MODULE

The Result Management Module is used to evaluate the answers submitted by the user. It compares the responses with correct answers, calculates the total score, and generates the final result. It also stores the result details and displays the user's performance after quiz completion. This module also stores the score details for future reference and displays the performance summary clearly to the user. It helps in quick and accurate result generation. It provides instant feedback after quiz completion, allowing users to know their performance immediately. The module can display details such as total questions, number of correct answers, wrong answers, and total score obtained. It helps users identify their strengths and weaknesses for further improvement.

This module also maintains result records in an organized manner, making it easy to retrieve previous scores whenever needed. It reduces manual effort in result preparation and ensures fairness in evaluation. Overall, the Result Management Module plays an important role in making the Online Quiz System efficient, reliable, and user-friendly.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 CONCLUSION

The Online Quiz System is an effective and user-friendly application developed to modernize the traditional quiz process. It provides a digital platform where users can attend quizzes easily and receive results instantly. The system reduces the time and effort required for manual paper-based examinations and improves the overall efficiency of conducting tests.

This project enures evaluation by automatically checking answers and calculating scores without human error. It also creates a smooth experience for users through a simple interface that allows easy navigation, question answering, and result viewing. Such features make the system suitable for students, teachers, and educational institutions.

6.2 FUTURE SCOPE

The Online Quiz System has a wide future scope for further development and improvement. Additional features such as timer-based quizzes, random question generation, and negative marking can be added to make the system more advanced and similar to real competitive examinations. These enhancements will improve the testing experience and make the system more effective.

The project can also be upgraded by integrating a database system to store user details, question banks, scores, and result history permanently. Features like user registration, login accounts, performance tracking, and progress reports can help users monitor their improvement over time. This will make the system more organized and useful for long-term use.

APPENDICES

APPENDIX A-SOURCE CODE

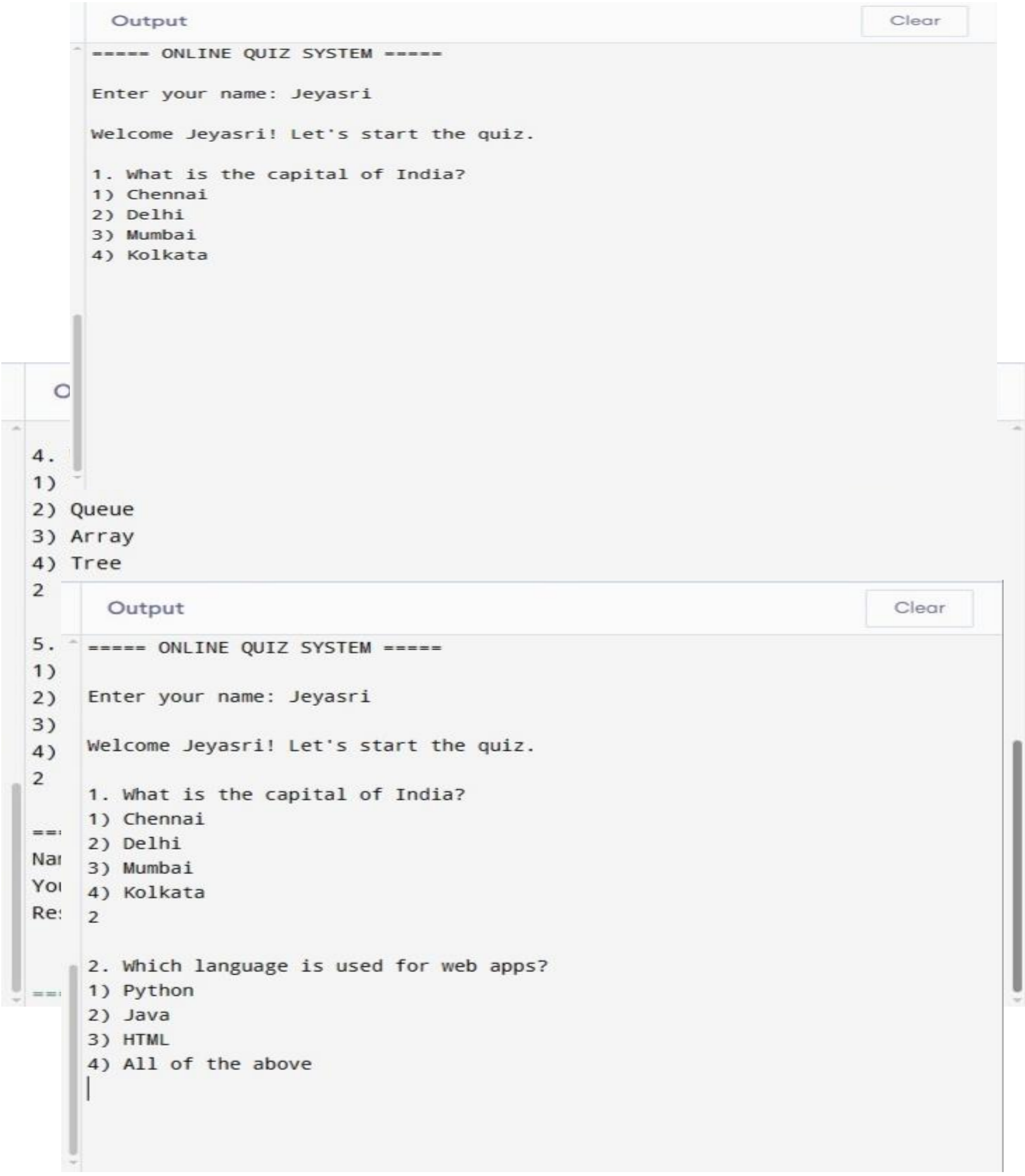
```
#include <stdio.h>


int main() {
    int score = 0;
    int answer;
    char name[50];


    printf("===== ONLINE QUIZ
SYSTEM =====\n\n");


    // Get user name
    printf("Enter your name: ");
    scanf("%s", name);


    printf("\nWelcome %s! Let's start
the quiz.\n\n", name);


    // Question 1
    printf("1. What is the capital of
India?\n");
```

```
    printf("1) Chennai\n2) Delhi\n3)
Mumbai\n4) Kolkata\n");
    scanf("%d", &answer);
    if (answer == 2) score++;

    // Question 2
    printf("\n2. Which language is used
for web apps?\n");
    printf("1) Python\n2) Java\n3)
HTML\n4) All of the above\n");
    scanf("%d", &answer);
    if (answer == 4) score++;

    // Question 3
    printf("\n3. Which operator is used
for modulus in C?\n");
    printf("1) /\n2) %%\n3) *\n4) +\n");
    scanf("%d", &answer);
    if (answer == 2) score++;

    // Question 4
    printf("\n4. Which data structure
uses FIFO?\n");
    printf("1) Stack\n2) Queue\n3)
```

```
Array\n4) Tree\n");
    scanf("%d", &answer);
    if (answer == 2) score++;

    // Question 5
    printf("\n5. Which keyword is used
to return a value?\n");
    printf("1) break\n2) return\n3)
exit\n4) stop\n");
    scanf("%d", &answer);
    if (answer == 2) score++;

    // Result
    printf("\n===== RESULT
=====\n");
    printf("Name: %s\n", name);
    printf("Your Score: %d / 5\n",
score);

    if (score >= 3) {
        printf("Result: Pass\n");
    } else {
        printf("Result: Fail\n");
    }
```

```
    return 0;
}
```

RESULTS



Output Clear

```
4) All of the above
1

3. Which operator is used for modulus in C?
1) /
2) %
3) *
4) +
2

4. Which data structure uses FIFO?
1) Stack
2) Queue
3) Array
4) Tree
2

5. Which keyword is used to return a value?
1) break
2) return
3) exit
4) stop
```

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